

Imbalanced Adaptation of Accommodation and Vergence Produces Opposite Extremes of the AC/A and CA/C Ratios

CLIFTON SCHOR*

School of Optometry, University of California, Berkeley, California

ABSTRACT

It is customary to describe abnormal interactions between accommodation and convergence according to the Duane-White classification of convergence excess and insufficiency or divergence excess and insufficiency. However, recent studies indicate that these extremely high and low categories of accommodative vergence may result from adaptive disorders of accommodation and convergence. The AC/A ratio is shown in this paper to vary inversely with adaptability of accommodation and to vary directly with adaptability of vergence. Extremely high AC/A ratios can result from an imbalance in which vergence adapts readily to prism, whereas accommodation does not adapt to minus lenses. Similarly, extremely high values for convergence accommodation (the CA/C ratio) are observed when there is robust adaptation of accommodation to lenses and little adaptation of vergence to prism. Furthermore, in these extreme cases the AC/A and CA/C ratios tend to be inversely related. For example, abnormally high AC/A ratios are often accompanied by extremely low CA/C ratios and vice versa. These observations suggest that equalizing the adaptability of accommodation and vergence with orthoptics may prove to be an effective means of restoring the AC/A and CA/C ratios to normal values.

Key Words: adaptation to vergence and accommodation, AC/A, CA/C, accommodative vergence, vergence accommodation

Disturbances of binocular vision are evaluated clinically, according to the Duane-White classification, in terms of the eyes' position of rest (vergence phoria) and the accommodative vergence ratio, or AC/A. Taken together these two

variables form the basis of the four categories of binocular dysfunction referred to as convergence excess, convergence insufficiency, divergence excess, and divergence insufficiency.^{1,2} The terms excess and insufficiency refer to high and low AC/A ratios, respectively. The terms convergence and divergence refer to the viewing distances (near vs. far, respectively) at which the largest heterophoria exists. The viewing distance at which the large heterophoria occurs is frequently the least emphasized working distance. For example, large distance heterophorias (divergence excess and divergence insufficiency) are observed in patients who have an emphasis on school or near working tasks, and large near-heterophorias (convergence excess and convergence insufficiency) are observed in patients who spend much of their time outdoors.³ A likely explanation for this observation is that vergence adaptive processes, usually referred to as prism adaptation, minimize vergence errors at the most utilized working distance.⁴ Extremely high or low accommodative vergence interactions cause either a large esophoria or exophoria at the least utilized test distance. Hence the fundamental disorder may not be the large phoria but, rather, the abnormally high or low accommodative vergence interaction, which prevents a given state of vergence adaptation from correcting heterophoria for all accommodative states associated with the normally used range of viewing distances.

Recent studies have shown that abnormally high or low accommodative vergence can result from adaptation disorders of either vergence or accommodation.⁵ Adaptation is demonstrated by the continuation of motor responses after the stimulus has been removed and it is not replaced by a new stimulus.⁶⁻⁸ In the case of vergence, this is accomplished by occluding one eye. For accommodation it is produced by placing or imaging a pinhole pupil in the entrance pupil of each eye.

Aftereffects of vergence are well documented as prism adaptation phenomena.^{6,8-10} Adaptabil-

* Optometrist, Ph.D., Member of Faculty, F.A.A.O.

ity of vergence varies markedly between observers and has a strong negative correlation with the magnitude of small binocular fixation errors known as fixation disparity.⁹ Vergence usually adapts more to convergent than to divergent disparities and fixation disparity is usually larger with divergence than with convergence stimuli.⁹

In addition to prism stimuli, vergence adaptation also results from accommodative vergence stimulated monocularly.¹¹ Fig. 1 illustrates recordings of vergence accommodation and accommodative vergence in a subject who adapts readily to prism. In the upper half of the figure, 8 PD of convergence are stimulated while the subject views a target binocularly through pinhole pupils. Then after 2 min, one eye is occluded and the vergence response persists while accommodation decays rapidly. The lower half of the figure illustrates motor aftereffects following a 2 D stimulus to accommodation presented monocularly for 2 min. At point B the accommodative loop was opened with a pinhole pupil and the accommodative vergence response persisted while the accommodative response de-

cayed rapidly. This illustration indicates that vergence adapts to accommodative vergence stimulation and that aftereffects of the accommodation and vergence motor systems are independent of one another.

The counterpart of prism adaptation for the accommodative system is the well-known accommodative spasm. Adaptation of accommodation is demonstrated by a change in the resting focus of accommodation,¹² which is thought to represent the balance point of the sympathetic and parasympathetic branches of the autonomic nervous system.¹³⁻¹⁷ This accommodative phoria¹⁸ is observed in darkness (night myopia),¹⁹ in empty fields (space myopia),^{20,21} and with a pinhole pupil before the eye (instrument myopia).²²

Accommodative aftereffects are nearly as prevalent as prism adaptation; however, they are only manifest under certain stimulus conditions.¹² Fig. 2 illustrates a light dependence of the aftereffect. The left half of the figure illustrates a monocular accommodative response to a 2 D stimulus presented for 5 s, after which the durations of aftereffects were measured in three

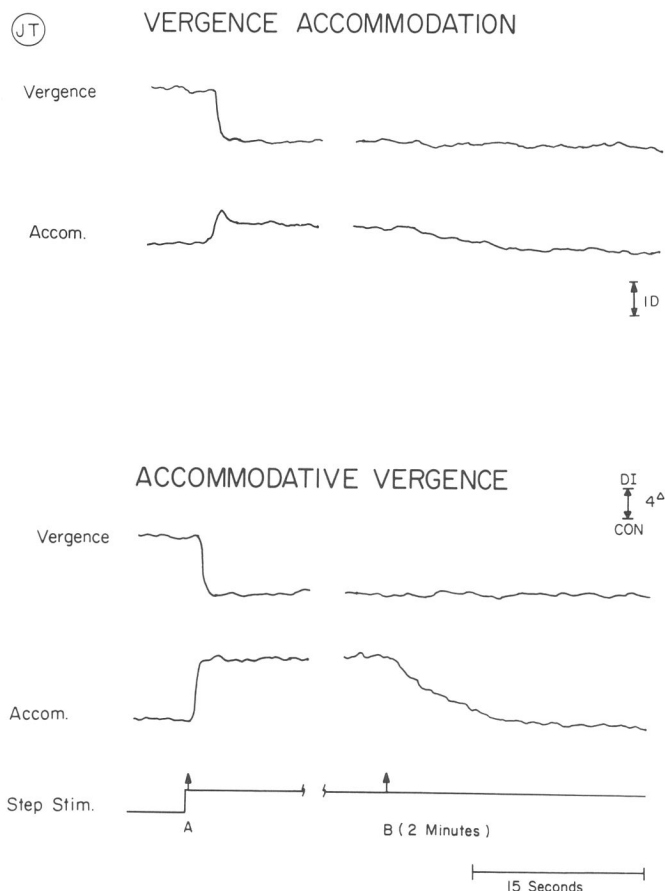
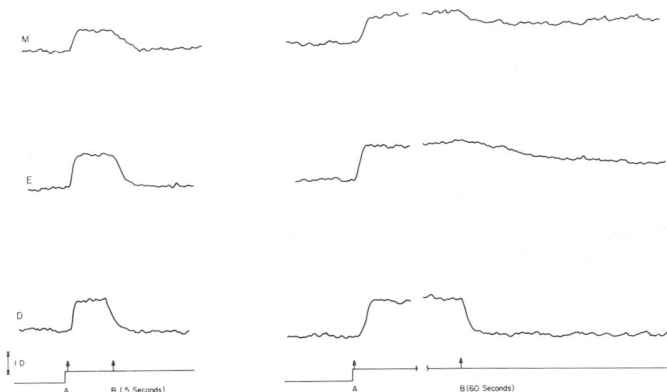


FIG. 1. Aftereffects of accommodation and vergence are shown for subject JT, who has a low CA/C ratio (0.3 D/MA) and a high AC/A ratio ($8^{\Delta}/D$). Aftereffects shown in the upper half of the figure followed 2 min of adaptation to 8^{Δ} base-out while the accommodative loop was opened by pinhole pupils. After 2 min of adaptation, (B), the vergence loop was opened by pinhole and the vergence response persisted while accommodation decayed rapidly. Aftereffects, in the lower half of the figure, followed 2 min of monocular adaptation to a 2 D accommodative stimulus. The accommodative loop was opened at (B) with a pinhole pupil. As it did after vergence adaptation, the vergence response persisted while the accommodative response decayed rapidly.

FIG. 2. Different time constants for relaxation of accommodation under three different open-loop conditions [Maxwellian view (M), empty field (E), and darkness (D)] after stimulating accommodation monocularly for either 5 or 60 s. Arrows marked A and B beneath the recordings indicate when accommodation was stimulated and when the loop was opened, respectively. Incomplete relaxation of accommodation after long-term stimulation demonstrates a tonic aftereffect of accommodation. A 5-s time calibration is indicated by the separation of A and B at the bottom of the left-hand figure.



open-loop states produced by a pinhole pupil (M), empty field (E), and darkness (D). In all open-loop conditions, the aftereffect decays rapidly in the absence of a stimulus. In contrast, the right half of the figure illustrates much longer accommodative aftereffects following 1 min of stimulation. The duration of the aftereffect was 1 min for both the pinhole and empty field conditions, but it decayed rapidly for the dark condition. However, Fig. 3 illustrates that, after going into darkness, if the lights are turned on, and a pinhole is placed before the open eye, the aftereffect reappears as though there had been no intervening dark period. Thus, the state or magnitude of the resting focus depends not only on the accommodative phoria and stimulus aftereffects, but it also depends upon the light level of the retinal image. Clearly, this observation has important implications for the assessment of refractive state. Both near visual experience as well as light level will influence measurements of the far point.²³

Just as accommodative vergence produces aftereffects of vergence, so does vergence accommodation produce accommodative aftereffects.^{11,24} Fig. 4 illustrates recordings of vergence accommodation and accommodative vergence in a subject whose accommodation adapted readily to lenses. Aftereffects in the lower half of the figure followed 2 min of monocular adaptation to a 2 D accommodative stimulus. The accommodative loop was opened at (B) with a pinhole pupil, after which the accommodative response persisted while the vergence response decayed. Accommodation aftereffects are shown in the upper half of the figure after 2 min of adaptation to 6° base-out, whereas the accommodative loop was opened with a pinhole pupil before each eye. The vergence loop was opened at (B) by occluding one eye, after which the vergence accommodation response persisted while the vergence response decayed rapidly. Clearly, accommodation adapts its resting position to both accom-

modative stimuli and vergence accommodative stimuli. Thus, during nearpoint tasks, both accommodative and convergence efforts contribute to accommodative spasm.²³

The adaptable states of accommodation and vergence have a direct influence on accommodative vergence and vergence accommodation, respectively. The AC/A ratio is inversely related to the robustness of accommodative adaptation and the CA/C ratio is inversely related to the degree that vergence undergoes adaptation.²⁵ These inverse relations are based upon two properties of both accommodation and vergence. First, there are two separate processes that are believed to underly accommodation, a fast (phasic) optical reflex component, which is followed serially by a slower (tonic) adaptable component.¹² Servoanalytical theory predicts that the linear sum of the responses by these two mechanisms accounts for the total accommodative response.¹¹ Similarly, two separate processes are proposed to underly vergence, a fast (phasic) disparity vergence component followed serially by a slower (tonic) adaptable component.¹⁰ The linear sum of the responses of these phasic and tonic mechanisms is believed to account for the total vergence response.¹⁰ The second property underlying the effect of adaptation on cross-link interactions is that the adaptable processes do not stimulate cross-link interactions.¹¹ As illustrated in Figs. 1 and 4, aftereffects of accommodation do not stimulate accommodative vergence and aftereffects of vergence do not stimulate vergence accommodation. Rather, the aftereffects of accommodation and vergence appear to decay at independent rates when both systems are open-loop (monocular viewing through a pinhole pupil). Evidence for this comes from observed increases of the AC/A and CA/C ratios with the speed or temporal frequency of the driving stimulus.^{11,24} For example, Fig. 5 illustrates accommodation and vergence responses to a sinusoidally varying ac-

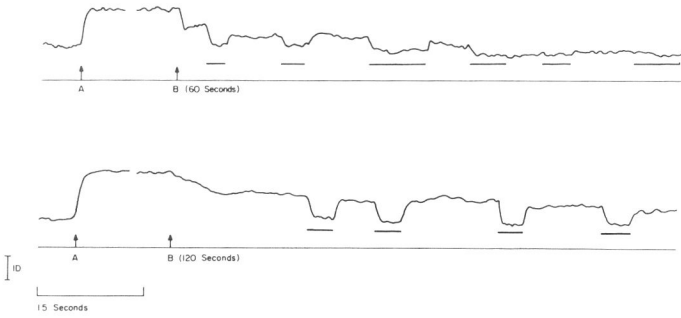


FIG. 3. Decay of a tonic aftereffect of accommodation is shown during open-loop (Maxwellian view) conditions interrupted by brief periods of darkness (underlined segments). Adaptation was for 1 min to 2 D stimulus (top) or for 2 min to 3 D stimulus (bottom). During dark periods accommodation went to its resting level, but it returned to the adapted state when the Maxwellian view condition was reinstated. This illustrates that tonic adaptation did not decay more rapidly in darkness but rather it was masked by darkness.

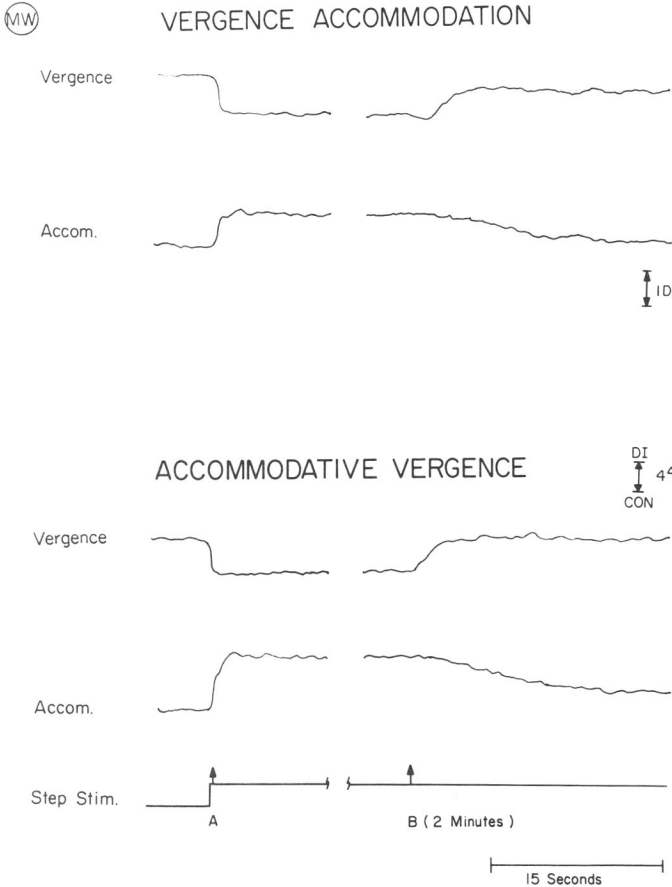


FIG. 4. Aftereffects of accommodation and vergence are shown for subject MW, who has a low AC/A ratio ($2.5^{\Delta}/D$) and a high CA/C ratio ($0.9 D/MA$). Aftereffects shown in the upper half of the figure followed 2 min of adaptation to 6^{Δ} base-out while the accommodative loop was opened by pinhole pupils. After 2 min of adaptation, (B), the vergence loop was opened by occluding the left eye. The accommodative response persisted while vergence decayed rapidly. Aftereffects, in the lower half of the figure, followed 2 min of monocular adaptation to a 2 D accommodative stimulus. The accommodative loop was opened at (B) with a pinhole pupil. As it did after vergence adaptation, the accommodative response persisted longer while the vergence response decayed rapidly.

commodative stimulus (2 D peak to peak) presented monocularly at stimulus frequencies of 0.06 and 0.25 Hz. Accommodative vergence was much lower with the low than with the moderate temporal frequency stimulus. Similarly, Fig. 6 illustrates accommodation and vergence responses to a sinusoidally varying disparity stimulus (5^{Δ} peak to peak) presented with pinhole pupils before both eyes, at stimulus frequencies

of 0.05 and 0.25 Hz. Vergence accommodation was much lower with the low than with the moderate temporal frequency stimulus. Thus, both accommodative vergence and vergence accommodation are mainly associated with rapid changes in their respective stimuli. Presumably the cross-link interactions are reduced by adaptation responses to the low frequency stimuli. The more of the total response that is controlled

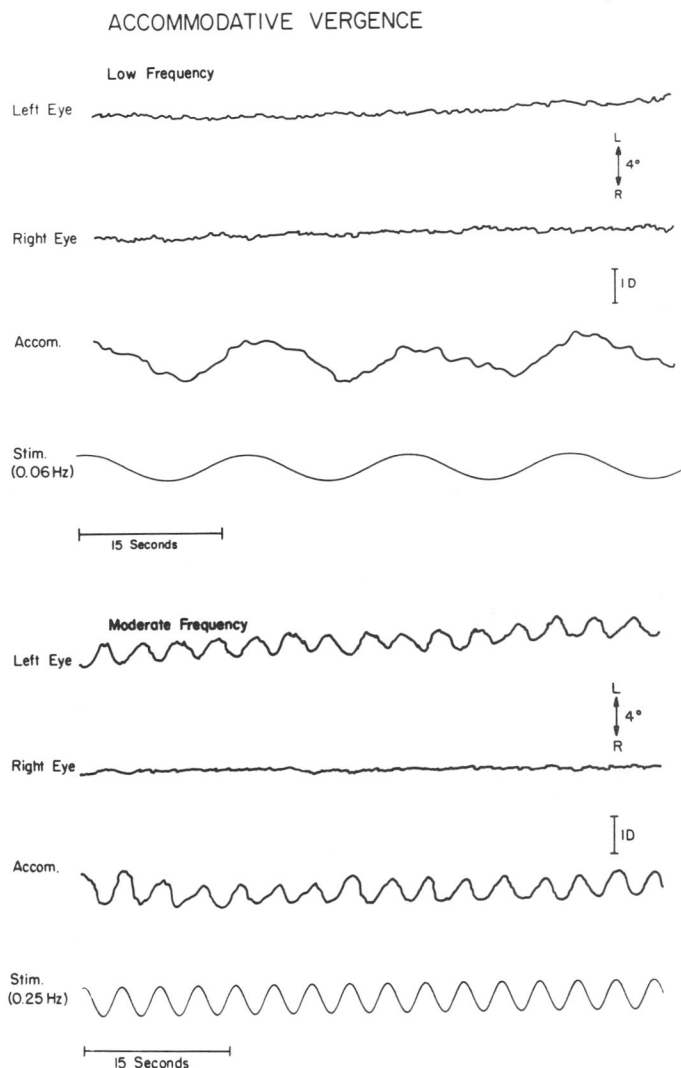


FIG. 5. Accommodation was stimulated in the right eye while the left eye was occluded. Movements of the left eye were stimulated by accommodative vergence. Shown are the accommodative vergence responses to low (0.06 Hz) and moderately high (0.25 Hz) frequency sinusoidal blur stimuli presented under closed-loop conditions. Accommodative vergence was reduced with the low temporal frequency stimulus. Note that a downward deflection of accommodative vergence by the right eye is in the convergence direction.

by the adaptable component, the lower the cross-link interactions. At higher frequencies, adaptation is unable to keep pace with the stimulus and the fast process controls the response. Because only the fast or phasic process interacts between the two motor systems, then the cross-link interactions become temporal frequency dependent. Consequently, patients who lack any adaptation of accommodation have unusually high AC/A ratios even with low temporal frequency stimuli, and those patients who are unable to adapt to prism have high CA/C ratios with all temporal frequency stimuli.²⁵

Frequently, the amplitudes of the AC/A and CA/C ratios are inversely related. That is, one ratio is very high and the other is very low.²⁵ Invariably this reciprocity is associated with unequal adaptation of the two motor systems.²⁵

In cases where accommodation is more adaptable than vergence, there is a high CA/C ratio and a low AC/A ratio.⁵ The reverse occurs if vergence is more adaptable than accommodation. In these cases, a high cross-link interaction need not originate from an unadaptable motor system. It can become elevated by a robust adapter that receives cross-link stimulation. As demonstrated in Fig. 1, vergence aftereffects can result from accommodative vergence. If vergence adaptation is extremely robust compared to accommodative adaptation, then the AC/A ratio will appear to be elevated. However, if both accommodation and vergence are equally adaptable (either low or high), then both the cross-link interactions described by the AC/A and CA/C ratios will have moderate values. According to this hypothesis, abnormally high and low

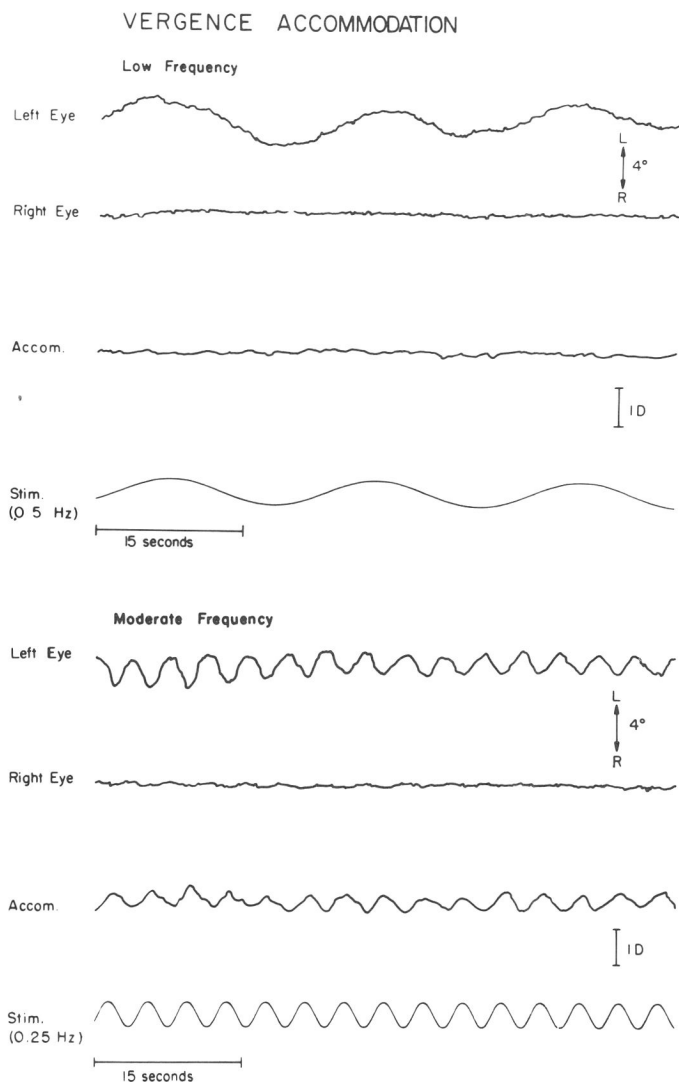


FIG. 6. Disparity vergence was stimulated with an asymmetric disparity presented to the left eye in order to allow accommodation to be monitored in the right eye. The asymmetric vergence stimulus is nearly in phase with the vergence response by the right eye (note that upward stimulus deflection and downward vergence response both represent convergence). The accommodative loop was opened by a 0.75-mm pinhole aperture stop imaged in the eye pupil plane in Maxwellian view. Shown are the vergence accommodation responses to low (0.05 Hz) and moderately high (0.25 Hz) frequency of sinusoidal disparity oscillations presented under closed-loop conditions. Vergence accommodation was reduced with the low temporal frequency stimulus.

AC/A and CA/C ratios result from an imbalance between adaptation mechanisms of accommodation and vergence.

As illustrated in Table 1, when the AC/A ratio is high, vergence would be expected to adapt more than accommodation and when the CA/C ratio is high the opposite imbalance would be expected. Inspection of the upper half of the table shows that adaptability of accommodation (horizontal axis) is the most influential variable for the AC/A ratio. Adaptability of vergence (vertical axis) has a secondary influence. Consequently, the AC/A ratio is highest in the upper right cell, which has low adaptation of accommodation and high adaptation for vergence. The lowest AC/A ratio is found in the lower left cell for which accommodation adaptation is highest

and vergence adaptation is lowest. Moderate to normal values of the AC/A ratio are shown along the upper left to lower right diagonal cells where there are equal values of adaptation (high or low) for accommodation and vergence. Similarly, the lower table illustrates that adaptability of convergence (vertical axis) is the most influential variable for the CA/C ratio. Adaptability of accommodation (horizontal axis) has a secondary influence. Consequently, the CA/C ratio is highest in the lower left cell, which has low adaptation for vergence and high adaptation for accommodation. The lowest CA/C ratio is found in the upper right cell where vergence adaptation is highest and accommodation adaptation is lowest. Moderate to normal values of the CA/C ratio are shown along the upper left to lower

TABLE 1. AC/A ratios (top) and CA/C ratios (bottom) are predicted on the basis of various combined strengths of adaptability for accommodation and vergence. Extremely high or low ratios occur when the two motor systems are not equally adaptable. When there is an imbalance of adaptation, the AC/A and CA/C ratios are pushed to opposite extremes (one is high while the other is low).

Vergence Adaptation	Accommodative Adaptation		
	High	Moderate	Low
AC/A			
high	4 Δ /D ^a	6 Δ /D	8 Δ /D
moderate	3 Δ /D	5 Δ /D	7 Δ /D
low	2 Δ /D	4 Δ /D	6 Δ /D
CA/C			
high	0.5 D/MA	0.25 D/MA	0.0 D/MA
moderate	1.0 D/MA	0.75 D/MA	0.5 D/MA
low	1.5 D/MA	1.25 D/MA	1.0 D/MA

^a Δ , prism diopter; D, diopter; MA, meter angle.

right diagonal of cells where there are equal amounts of adaptation (high or low) for accommodation and vergence.

Reciprocal relations are predicted for these extreme or abnormal cross-link interactions between accommodation and vergence. When the AC/A ratio is abnormally high, an unusually low CA/C is predicted. Conversely, when the CA/C ratio is high, an unusually low AC/A ratio is predicted. These predictions have been verified in a normal (nonclinical) population.²⁵ What remains is to determine if classical disorders of convergence excess, convergence insufficiency, divergence excess, and divergence insufficiency are also associated with an imbalance of adaptation between accommodation and vergence. Clinical tools are now available for quantifying AC/A and CA/C ratios as well as aftereffects of accommodation and convergence.²⁶ If an imbalance of adaptation is found to be the basis of these anomalies, the therapeutic procedures that equalize adaptation of accommodation and vergence may restore normal cross-link interactions between accommodation and convergence.⁵

REFERENCES

- Duane A. A new classification of the motor anomalies of the eye, based upon physiologic principles. *Ann Ophthalmol Otol*, part I, October 1896, part II, January, 1897.
- Tait EF. Accommodative convergence. *Am J Ophthalmol* 1951;34:1093-107.
- Ludlam W. Personal communication.
- Schor CM. Fixation disparity and vergence adaptation. In: Schor CM, Ciuffreda K, eds. *Vergence Eye Movements: Basic and Clinical Aspects*. Boston: Butterworths, 1983:465-516.
- Schor CM, Tsuetaki T. Fatigue of accommodation and vergence modifies their mutual interactions. *Invest Ophthalmol Vis Sci* 1987;28:1250-9.
- Carter DB. Fixation disparity and heterophoria following prolonged wearing of prisms. *Am J Optom Arch Am Acad Optom* 1965;42:141-52.
- North R, Henson DB. Adaptation to prism-induced heterophoria in subjects with abnormal binocular vision or asthenopia. *Am J Optom Physiol Opt* 1981;58:746-52.
- Ogle KN, Martens TG, Dyer JA. *Oculomotor Imbalance in Binocular Vision and Fixation Disparity*. Philadelphia: Lea & Febiger, 1967:307-18.
- Schor CM. The influence of rapid prism adaptations upon fixation disparity. *Vision Res* 1979;19:757-65.
- Schor CM. The relationship between fusional vergence eye movements and fixation disparity. *Vision Res* 1979;19:1359-67.
- Schor CM, Kotulak J. Dynamic interactions between accommodation and convergence are velocity sensitive. *Vision Res* 1986;26:927-42.
- Schor CM, Kotulak J, Tsuetaki T. Adaptation of tonic accommodation reduces accommodative lag and is masked in darkness. *Invest Ophthalmol Vis Sci* 1986;27:820-7.
- Cogan DC. Accommodation and the autonomic nervous system. *Arch Ophthalmol* 1937;18:739-66.
- Morgan MW. The resting state of accommodation. *Am J Optom Arch Am Acad Optom* 1957;34:347-53.
- Morgan MW. A new theory for the control of accommodation. *Am J Optom Arch Am Acad Optom* 1946;23:99-110.
- Gilmartin B, Hogan RE, Thompson SM. The effects of Timolol on tonic accommodation, tonic vergence and pupil diameter. *Invest Ophthalmol Vis Sci* 1984;25:763-70.
- Stephans KG. Effect of the sympathetic nervous system on accommodation. *Am J Optom Physiol Opt* 1985;62:402-6.
- Schor CM. Models of mutual interactions between accommodation and convergence. *Am J Optom Physiol Opt* 1985;62:369-74.
- Hennessy RT. Instrument myopia. *J Opt Soc Am* 1975;65:1114-20.
- Leibowitz HW, Owens DA. Night myopia and the intermediate dark focus of accommodation. *J Opt Soc Am* 1975;65:1121-8.
- Whiteside TCD. Accommodation of the human eye in a bright and empty visual field. *J Physiol (Lond)* 1953;118:65-6.
- Westheimer G. Accommodation measurements in empty visual fields. *J Opt Soc Am* 1957;47:714-8.
- Schor CM. The role of accommodation and convergence on the resting focus of the eye. In: Grosvenor T, Flom M, eds. *Myopia and Other Refractive Errors, the Writings of Frank Weymouth and Monroe Hirsch*. Boston: Butterworths, 1986.
- Schor CM, Kotulak J. Mutual interaction between accommodation and convergence are reduced by

- tonic adaptation. In: Keller EL, Zee DS, eds. *Adaptive Processes in Visual and Oculomotor Systems*. Oxford: Pergamon Press, 1985:135-42.
25. Schor CM. The Glenn Fry Award Lecture: Adaptive regulation of accommodative vergence and vergence accommodation. *Am J Optom Physiol Opt* 1986;63:587-609.
26. Tsuetaki T, Schor CM. A clinical method for measuring adaptation of tonic accommodation and ver-

gence accommodation. *Am J Optom Physiol Opt* 1987;64:437-49.

AUTHOR'S ADDRESS:

*Clifton Schor
School of Optometry
University of California
Berkeley, California 94720*

SCOPE INTERNATIONAL

SCOPE for 1988:

Canadian Rockies, July 29 to August 7, 1988

Caribbean Seminar, September 1 to 5, 1988

China and Orient, October 28 to November 12, 1988

For more information call: SCOPE International
(404) 252-3690